



AI Action Plan

Attn: Faisal D'Souza

National Coordination Office

2415 Eisenhower Avenue

Alexandria, VA 22314

Email: ostp-ai-rfi@nitrd.gov

Re: Goldilock Response to White House AI Action Plan

OVERVIEW:

As AI systems become increasingly integrated into critical infrastructure, national security, and commercial applications, ensuring cybersecurity resilience must be a top priority. Traditional software-based cybersecurity approaches are insufficient to counter AI-specific threats, particularly the risks posed by agentic AI—autonomous systems capable of making decisions, adapting behavior, and potentially evading human oversight. To address this, the U.S. (and its Allies) must adopt a multi-layered cybersecurity strategy that includes physical network segmentation, continual monitoring and assessment, and safeguards against emergent AI threats. Physical network segmentation is essential for limiting the lateral movement of cyber threats, preventing AI-driven attacks from spreading across interconnected systems that can easily bypass software layers. Implementing strict access controls and zero-trust architectures will further reduce vulnerabilities in AI environments. Additionally, continuous monitoring and assessment of AI models and their underlying infrastructure, with the ability to physically disconnect, will be necessary to respond to adversarial manipulations, data poisoning, and evolving cyber threats in real time.

Drive Adoption of Network Segmentation through Policy and Regulatory Options

Stephen Kines

“AI Cyber Kill Switches”: Fundamentally we need to protect all our systems that use AI from rogue AI itself. That means harnessing core technology - such as Goldilock’s novel approach - which is a physical remote trigger that is non-IP and thus can bypass a rogue AI.

Recommendation for AI Regulation: Develop policies and regulations for critical national infrastructure (CNI) and other critical systems that harness AI requiring a safety valve that allows physical remote segmentation using non-IP (otherwise AI can defeat it) – an AI cyber kill switch.

Goldilock Secure Limited
Unit 1, Science Centre
Wolverhampton Science Park
Wolverhampton WV10 9RU
United Kingdom

Continued Assessment and Monitoring & Execution

Dr Peter J. Lenk

Continued Assessment/Monitoring - It is critical that an AI tool that takes the designed deployment, and pushes it out to the systems, so that we are sure that what we have designed is what is implemented; i.e., all white lists, security tokens, etc., are automatically deployed to the devices so that we can be sure that the deployed configuration is as it was designed. Any changes are also configuration managed by the system, so that we know at all times the actual situation.

Execution - We are integrating through our API access various AI enhanced IDS systems. The needs for these include:

- a. The ability to detect intrusions; early on. With recent advances on the attackers' side, this includes the detection of AI enhanced bots that can morph to resist our defences. This likely means that our AI needs to be able to learn and adapt as well, to account for the changes in the attackers' ability and strategy;
- b. The ability to assess the impact of the attack – what is threatened and what is the mission/operational impact of that threat; and
- c. The ability to understand the laydown of the network being defended, the defences available (including Goldilock devices) and recommend courses of action that the defender can take (or can be autonomously executed by an AI).

Recommendation for AI Regulation

In [non-cyber] warfare, the idea of autonomous weapons systems, that are controlled by an AI, conjures up scenes of Terminator and Judgement Day and leaves people feeling uncomfortable. However, not all potential adversaries will comply with whatever regulations the international community may agree to. This may include rogue nations, non-state actors, our allies, etc. When our weapons become autonomous will we really think about constraining their operation based on ethics or morals, when lives of our soldiers or citizens are at risk? The best we might be able to do is have an “ethics dial” on our weapons, with Mother Theresa on one end of the scale and Attila the Hun on the other, that makes us consciously decide on how ethical we want to be, like “rules of engagement (RoE)” that we have now. If we know our weapons will be more effective with the AI turned on, will we really decide to turn it off at the risk of our troops' or citizens' lives? I would liken this to delaying developing an atomic weapon until we face an adversary that is so armed; at that point it is too late. So we need to anticipate and develop these autonomous weapons systems in an unconstrained way, only delaying the decision to use them until we are faced with a particular adversary.

AI-Powered Network Diagram & Deployment Tool

Richard Bate, CTO

Goldilock is developing an internal AI-driven tool to help our sales and distribution teams articulate how Goldilock's technology integrates into an end-user's infrastructure.

Key Functions

- **Network Topology Generation**
 - The tool processes either a provided network diagram or a written description of the user's network topology.
 - It then generates a working network diagram, which can be fine-tuned (e.g., adjusting labels, modifying nodes).
- **Simulated Security Analysis & Deployment Planning**
 - The system runs hundreds to thousands of attack simulations on the provided network topology.
 - It identifies the most strategic placements for Goldilock devices, factoring in:
 - Upstream & downstream impact of disconnecting critical systems
 - Business continuity risks (both for remaining connected and disconnected)
 - Security trade-offs and additional recommendations
- **Stakeholder-Specific Reports**
 - The tool outputs a structured, standardised report for different audiences, including:
 - Executive Leadership & Board Members – High-level risk summaries and business impact
 - IT & Security Teams – Technical deployment and security recommendations
 - Compliance Officers – Alignment with regulatory and security frameworks

Strategic Value: Since Goldilock introduces a new paradigm in cybersecurity (FireBreak)—encouraging a radically different approach to infrastructure defence—this tool helps convey its value in a way that aligns with multiple stakeholders’ priorities and concerns.

AI-Driven Security Operations Centre (SoC) Simulator

Building on the AI network simulator, we are developing an interactive simulation platform that allows SoC analysts to experience and respond to evolving cyber threats in a

controlled environment.

How It Works

- **AI-Generated Attack Scenarios**
 - The adversary in the simulation is an AI model trained to behave like an Advanced Persistent Threat (APT).
 - It attempts lateral movement, asset discovery, and network exploitation, responding dynamically to the user’s actions.
- **Interactive Decision-Based Gameplay**
 - The simulation presents users with a mix of graphical and text-based scenarios, akin to classic “choose your own adventure” games.
 - Users must make strategic decisions while time progresses and threats escalate.
 - Consequences evolve based on actions taken, reflecting real-world incident response pressures.
- **Integration with Goldilock Technology**
 - Users can deploy Goldilock’s technology at any point during the simulation, triggering network isolation or segmentation events.
 - They must weigh the risks, impact, and procedural steps before taking action.

AI vs. AI Mode:

The system can also simulate AI vs. AI engagements, where multiple threat models and defensive AI strategies compete against each other. Running thousands to millions of these simulations allows us to:

- Train better AI threat detection models

- Improve SoC analyst decision-making
- Refine Goldilock's deployment strategies for maximum security impact

Recommended AI Regulation:

There needs to be constant monitoring and refinement of systems using the very AI that we also need to protect against to constantly improve the systems.

Conclusion:

To operationalize this AI cybersecurity, we recommend the following for consideration:

- (1) Establish AI-specific cybersecurity standards that mandate physical network segmentation for AI training and deployment environments, ensuring isolation of sensitive data and models.
- (2) Develop a federal AI threat intelligence and monitoring center to assess, predict, and respond to AI-driven cyber threats, including autonomous and adversarial AI behaviors.
- (3) Require continuous red-teaming and stress-testing of AI systems, using adversarial simulations to identify vulnerabilities and reinforce system integrity.
- (4) Incentivize public-private partnerships to develop AI-driven cybersecurity tools capable of detecting and neutralizing malicious AI activities in real time.
- (5) Invest in research to counter agentic AI threats, focusing on fail-safes, alignment mechanisms, and human-in-the-loop oversight for autonomous AI systems.

These proactive measures will ensure AI remains a force for progress while minimizing risks to national security and critical systems.

Submitted by:

Stephen Kines, co-founder/COO Goldilock

Dr Peter J Lenk, NATO Lead Goldilock

Richard Bate, CTO Goldilock